

PH102

After Mid Semester Exams

by
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Chapter-6

The Magnetic Field of a Magnetized Object

Below concepts covered in the previous lecture

- The Field of a Magnetized Object
- Physical Interpretation of Bound Currents
- The Magnetic Field Inside Matter
- The Auxiliary Field H
- Ampere's law in Magnetized Materials
- A Deceptive Parallel
- Boundary Conditions
- Linear and Nonlinear Media
- Magnetic Susceptibility and Permeability

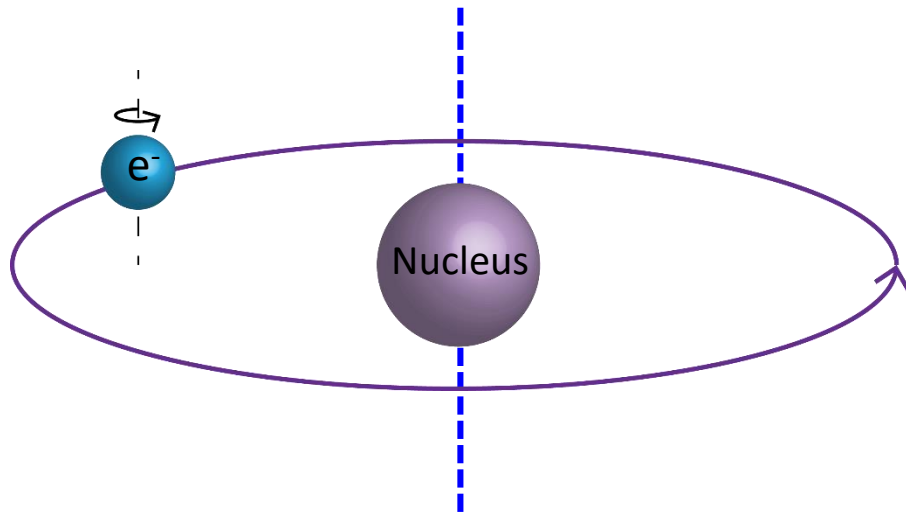
Below concepts covered in this lecture

- **Boundary Conditions**
- **Physics of Magnetism**
- **Linear and Nonlinear Media**
- **Magnetic Susceptibility and Permeability**
- **Faraday's Laws of Electromagnetic Induction**

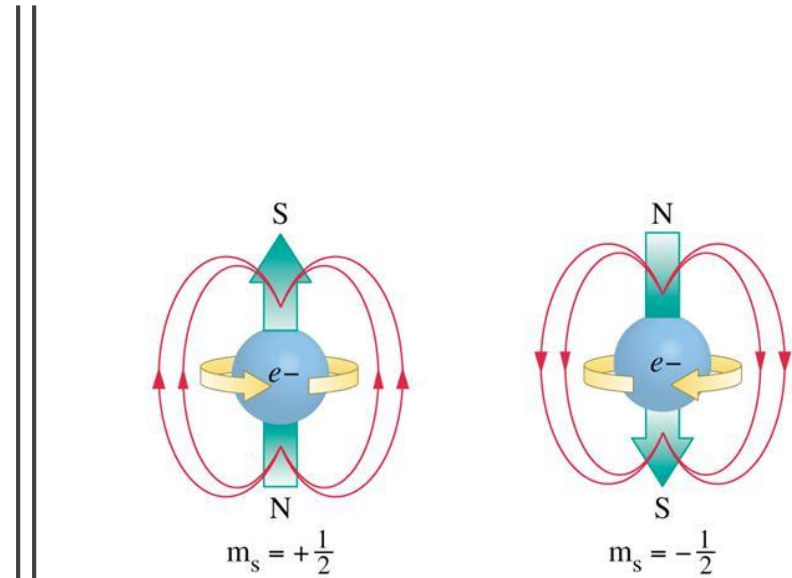
Physics of Magnetism

Magnetic phenomena are due to electric charges in motion/current-carrying wire

On an atomic scale the tiny currents i.e. electrons orbiting around nuclei and electrons spinning about their axes causing the Magnetic phenomena .



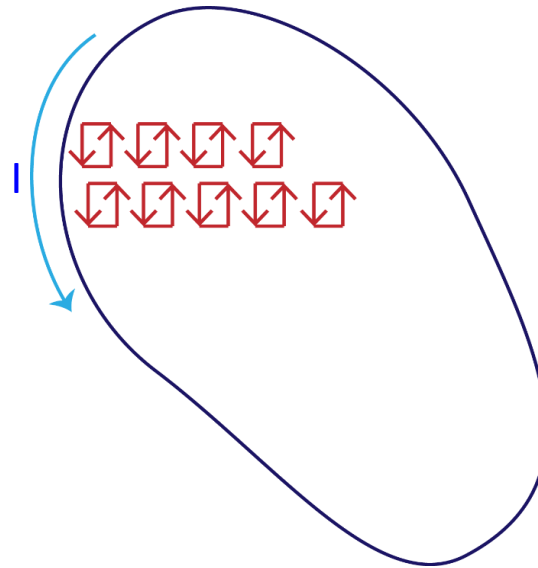
Electron orbiting around nuclei



Electron spinning around its axis

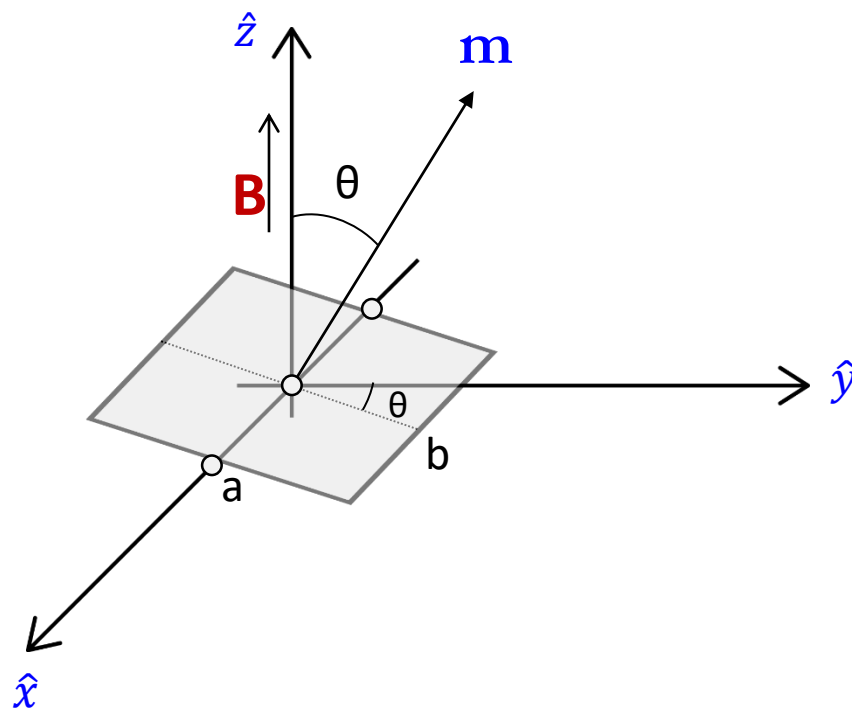
Torques and Forces on Magnetic Dipoles

A magnetic dipole experiences a torque in a magnetic field, just as an electric dipole does in an electric field.



Since any current loop could be built up from infinitesimal rectangles, with all the "internal" sides **canceling**, without any loss of generality.

Consider, a rectangular loop centered at the origin, and tilted at an angle θ from the z-axis towards the y-axis.



Let Magnetic Field $\mathbf{B}$ points along the z direction.

The forces on the two sloping sides cancel
(they tend to stretch the loop, but they don't rotate it).

The forces on the "horizontal" sides are likewise equal and opposite (so the net force on the loop is zero), but they do generate a torque:

$$N = aF \sin \theta \hat{x}$$

The magnitude of the force on each of these segments is

$$F = IbB$$

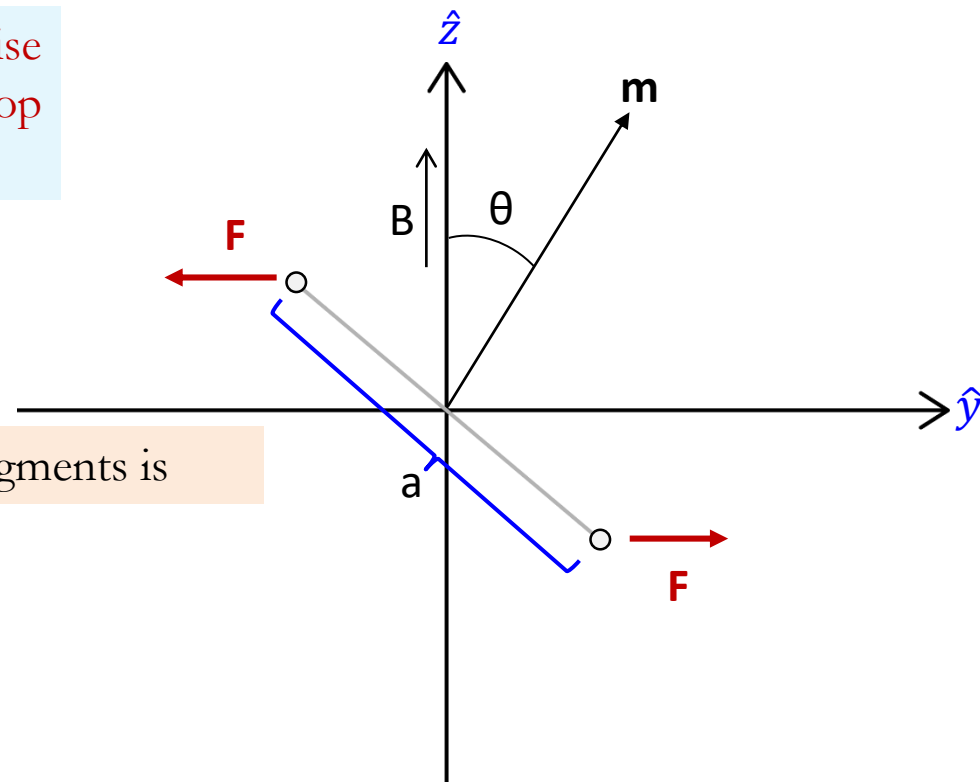
Hence

$$N = IabB \sin \theta \hat{x}$$

$$N = mB \sin \theta \hat{x}$$

Or

$$\vec{N} = \vec{m} \times \vec{B}$$



This equation gives the exact torque on any localized current distribution, in the presence of a uniform field.

The magnetic forces on a current carrying loop try to align the loop's normal vector (magnetic moment) with the magnetic field.

Example: Starting from the Lorentz force law show that the torque on any steady current distribution (not just a square loop) in a uniform field $\mathbf{B}$ is $\mathbf{m} \times \mathbf{B}$.

Solution

$$\overrightarrow{dF} = I (\overrightarrow{dl} \times \overrightarrow{B})$$

$$\overrightarrow{dN} = \vec{r} \times \overrightarrow{dF}$$

$$\overrightarrow{dN} = I [\vec{r} \times (\overrightarrow{dl} \times \overrightarrow{B})] \quad \text{Eq.1}$$

We know from vector triple product

$$(\vec{A} \times [\vec{B} \times \vec{C}]) + (\vec{B} \times [\vec{C} \times \vec{A}]) + (\vec{C} \times [\vec{B} \times \vec{A}]) = 0$$

$$(\vec{r} \times [\overrightarrow{dl} \times \overrightarrow{B}]) + (\overrightarrow{dl} \times [\overrightarrow{B} \times \vec{r}]) + (\overrightarrow{B} \times [\vec{r} \times \overrightarrow{dl}]) = 0 \quad \text{Eq.2}$$

But

$$d(\vec{r} \times [\vec{r} \times \overrightarrow{B}]) = d\vec{r} \times [\vec{r} \times \overrightarrow{B}] + \vec{r} \times [d\vec{r} \times \overrightarrow{B}]$$

Since $\mathbf{B}$ is constant

$$\overrightarrow{dr} = \overrightarrow{dl}$$

$$\overrightarrow{dr} = \overrightarrow{dl} \quad d(\vec{r} \times [\vec{r} \times \vec{B}]) = d\vec{l} \times [\vec{r} \times \vec{B}] + \vec{r} \times [\overrightarrow{dl} \times \vec{B}]$$

$$d(\vec{r} \times [\vec{r} \times \vec{B}]) = -d\vec{l} \times [\vec{B} \times \vec{r}] + \vec{r} \times [\overrightarrow{dl} \times \vec{B}]$$

$$\boxed{d\vec{l} \times [\vec{B} \times \vec{r}]} = \vec{r} \times [\overrightarrow{dl} \times \vec{B}] - d(\vec{r} \times [\vec{r} \times \vec{B}])$$

Substituting this value back to Eq. 2

$$(\vec{r} \times [\overrightarrow{dl} \times \vec{B}]) + \boxed{(\vec{r} \times [\overrightarrow{dl} \times \vec{B}] - d(\vec{r} \times [\vec{r} \times \vec{B}]))} + (\vec{B} \times [\vec{r} \times \overrightarrow{dl}]) = 0$$

$$2(\vec{r} \times [\overrightarrow{dl} \times \vec{B}]) - (d(\vec{r} \times [\vec{r} \times \vec{B}])) + (\vec{B} \times [\vec{r} \times \overrightarrow{dl}]) = 0$$

$$2(\vec{r} \times [\overrightarrow{dl} \times \vec{B}]) = (d(\vec{r} \times [\vec{r} \times \vec{B}])) - (\vec{B} \times [\vec{r} \times \overrightarrow{dl}])$$

Substituting this value back to Eq. 1

$$\boxed{\overrightarrow{dN} = I [\vec{r} \times (\overrightarrow{dl} \times \vec{B})]}$$

$$\overrightarrow{dN} = \frac{I}{2} \left[(d(\vec{r} \times [\vec{r} \times \vec{B}])) - (\vec{B} \times [\vec{r} \times \overrightarrow{dl}]) \right]$$

After taking the integration over a closed loop

$$\vec{N} = \frac{I}{2} \left[\left(\oint d(\underbrace{\vec{r} \times [\vec{r} \times \vec{B}]}_{\downarrow}) \right) - \left(\vec{B} \times \oint [\vec{r} \times d\vec{l}] \right) \right]$$

The first term of integration will become zero

$$\begin{aligned} \oint d f(r) \cdot dl &= f(r) \Big|_A^A \\ &= f(A) - f(A) = 0 \end{aligned}$$

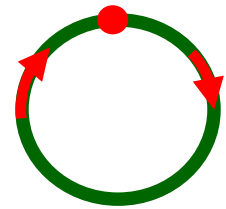
The second term will be

$$\oint [\vec{r} \times d\vec{l}] = 2a$$

$$\vec{N} = \frac{I}{2} [-(\vec{B} \times 2a)]$$

$$\vec{N} = -I [(\vec{B} \times \vec{a})]$$

$$\vec{N} = \vec{m} \times \vec{B}$$



Closed loop Integral

Diamagnets, Paramagnets and Ferromagnets

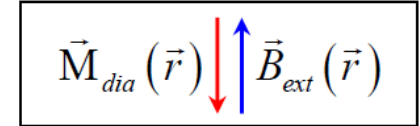
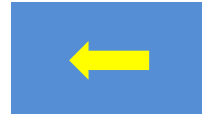
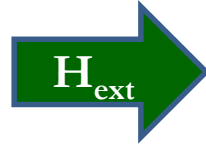
Normally, the total magnetic moment got canceled due to the random orientation of the atoms.

When a magnetic field is applied, a net alignment of these magnetic dipoles occurs, and the medium becomes magnetically polarized, or **Magnetized**.

Unlike electric polarization, which is almost always in the same direction as E , some materials acquire a magnetization parallel to B (paramagnets) and some opposite to B (Diamagnets).

Diamagnetism: Magnetic moment of filled shells of atoms. Induced moment opposes applied field.

Magnetic susceptibility χ_m of water at 20 °C -9.04×10^{-6} SI units

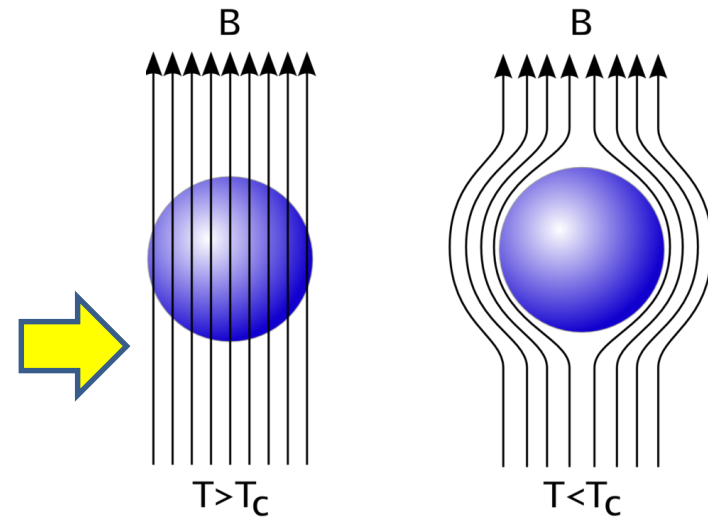


The most strongly diamagnetic material is Bismuth, $\chi = -1.66 \times 10^{-4}$

Diamagnets are materials with a magnetic permeability less than μ_0 (a relative permeability less than 1).

Magnetic susceptibility less than or equal to 0 (negative), since susceptibility is defined as $\chi_m = \frac{\mu}{\mu_0} - 1$

Transition from ordinary conductivity (left) to superconductivity (right). At the transition, the superconductor expels the magnetic field and then acts as a perfect diamagnet.



Paramagnetism: is a form of magnetism whereby certain materials are attracted by an externally applied magnetic field, and form internal, induced magnetic fields in the direction of the applied magnetic field.



H_{int} = Induced magnetic fields

Tungsten (W) $\chi = 6.8 \times 10^{-5}$

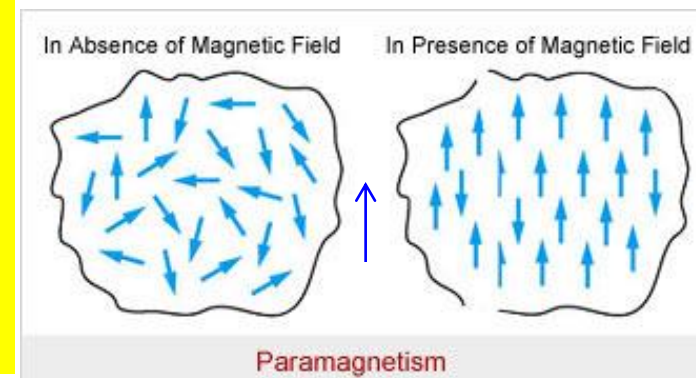
$$\vec{M}_{\text{para}}(\vec{r}) \uparrow \uparrow \vec{B}_{\text{ext}}(\vec{r})$$

Unfilled shells have a finite magnetic moment (orbital angular momentum) which aligns along the magnetic field direction.

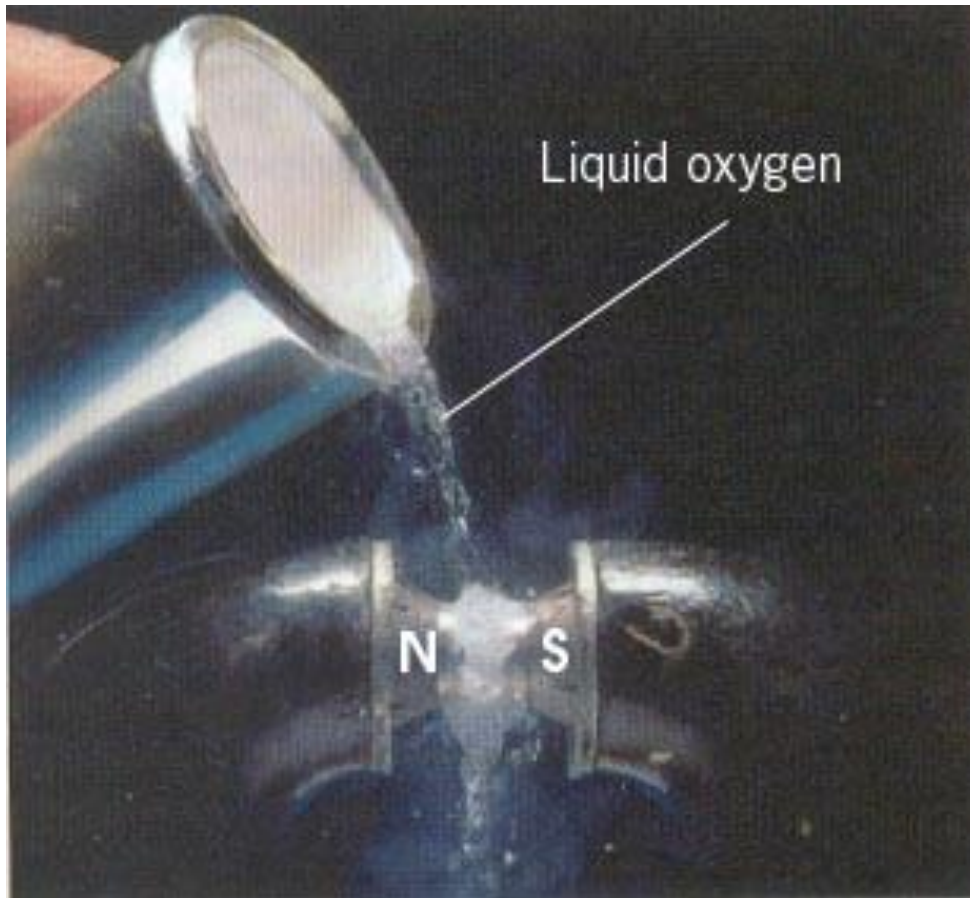
Curie's law $\Rightarrow M = \chi_m H = (C/T) H$

H_{int} = Induced magnetic fields

‘M’ is the resulting magnetization
 ‘ χ ’ is the magnetic susceptibility
 ‘H’ is the auxiliary magnetic field (Amp/meter)
 ‘T’ is absolute temperature, measured in kelvins
 ‘C’ is a material-specific Curie constant



Liquid oxygen is deflected by a magnetic field, illustrating its paramagnetism.

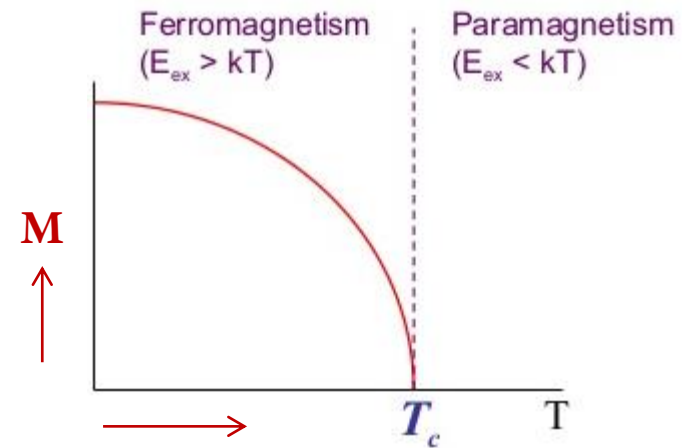
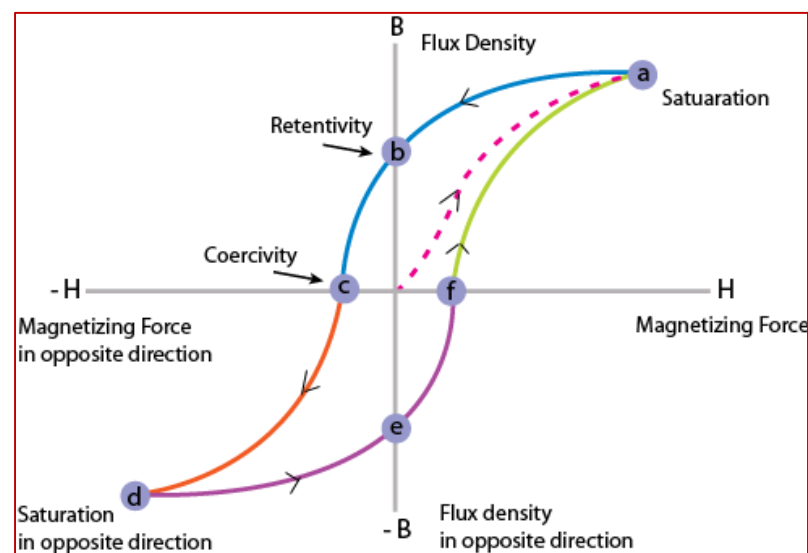
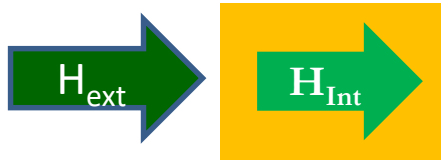


Paramagnetic materials are attracted to magnetic fields, hence have a relative magnetic permeability greater than one (or, equivalently, a positive magnetic susceptibility). The magnetic moment induced by the applied field is *linear* in the field strength and rather *weak*. It typically requires a sensitive analytical balance to detect the effect.

Ferromagnets

Some materials retain their magnetization even after the external field has been removed for these the magnetization is not determined by the present field but by the whole magnetic “History” of the object.

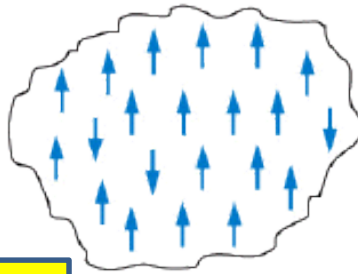
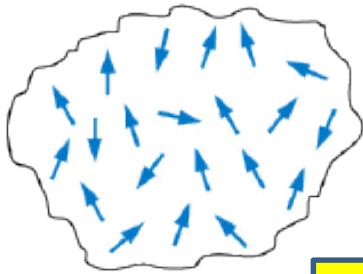
In Ferromagnetic crystals, the relationship between $\mathbf{M}$ and $\mathbf{H}$ is not linear. To accommodate this, a more general definition of differential susceptibility is used.



T_C = Ferromagnetic Cuire Temperature
Material Specific (e.g. Cobalt $T_C = 1388$ K)

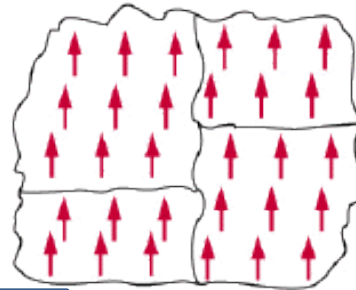
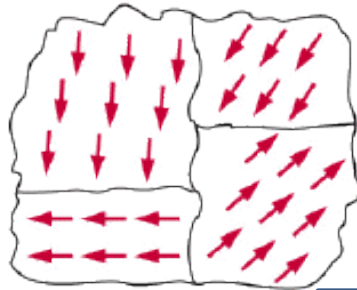
Magnetic field Absent

Presence of Magnetic field



Paramagnetism

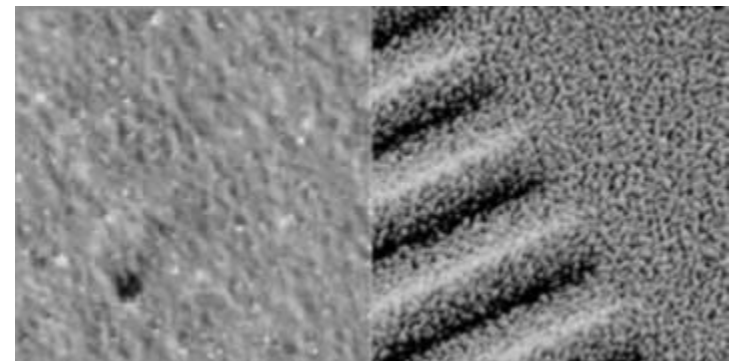
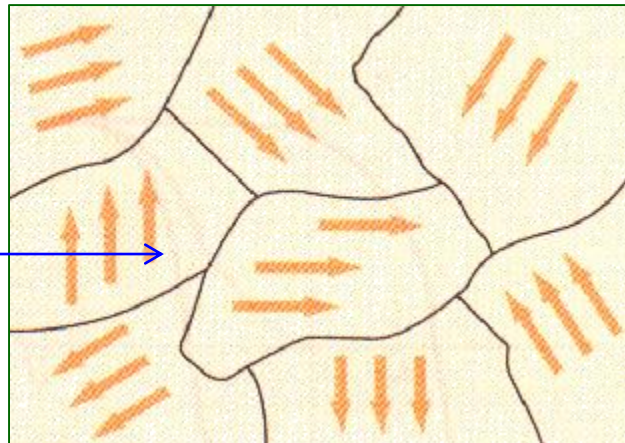
MFM detects magnetic field *gradients*



Ferromagnetism

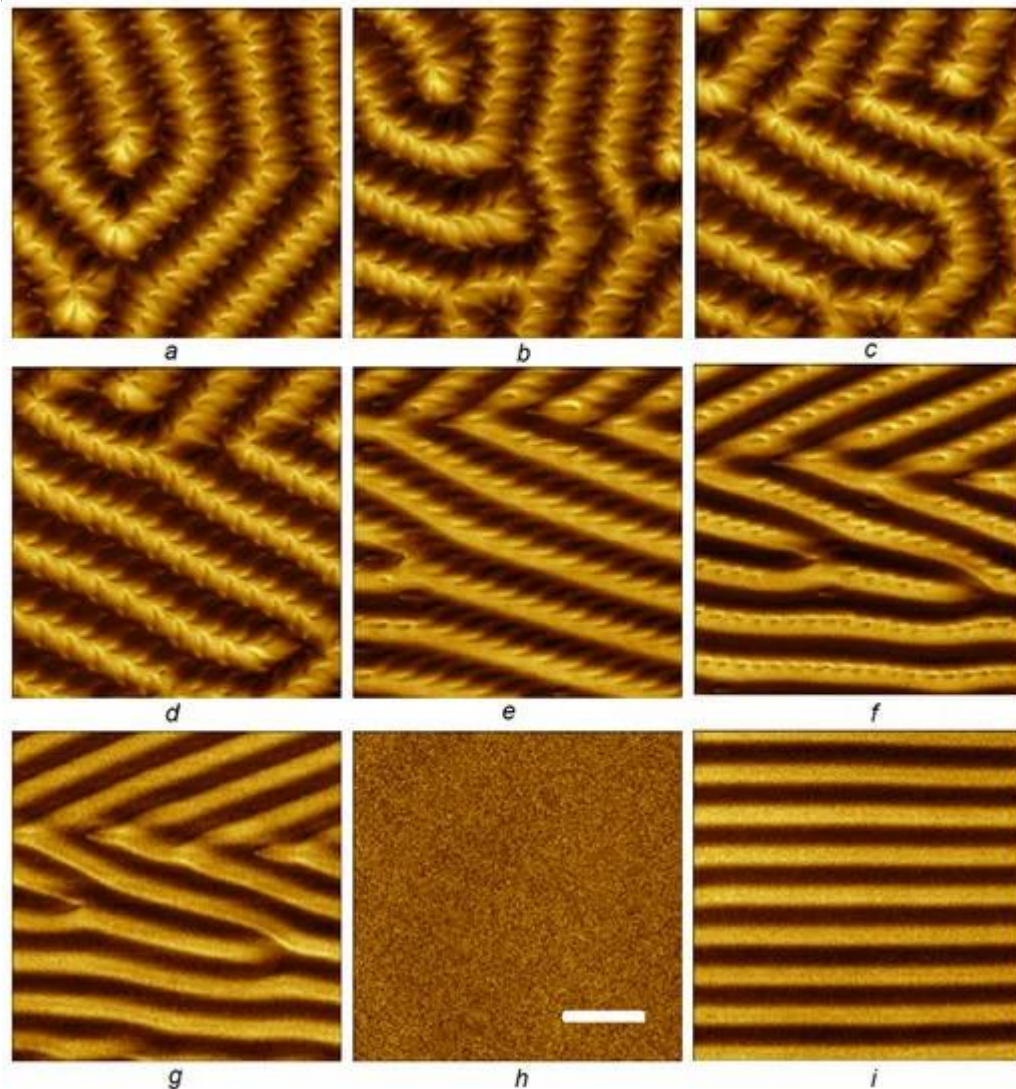
12 μm $\times$ 12 μm topographic (left) and magnetic force (right) images of a track edge on magnetic recording tape. The magnetic force image was taken at a lift height of 30 nm. The bright and dark regions in the magnetic force image correspond to boundaries between the magnetic bits

Magnetic Domains



Magnetic Force Microscopy

the MFM image is generated by measuring the amplitude and phase of cantilever oscillation. These signals contain information about magnetic domain distributions of the sample surface. MFM can be used to image naturally occurring or deliberately written domain structures in magnetic materials.

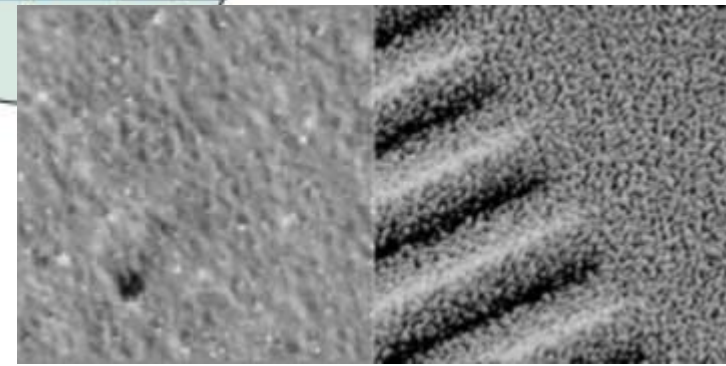
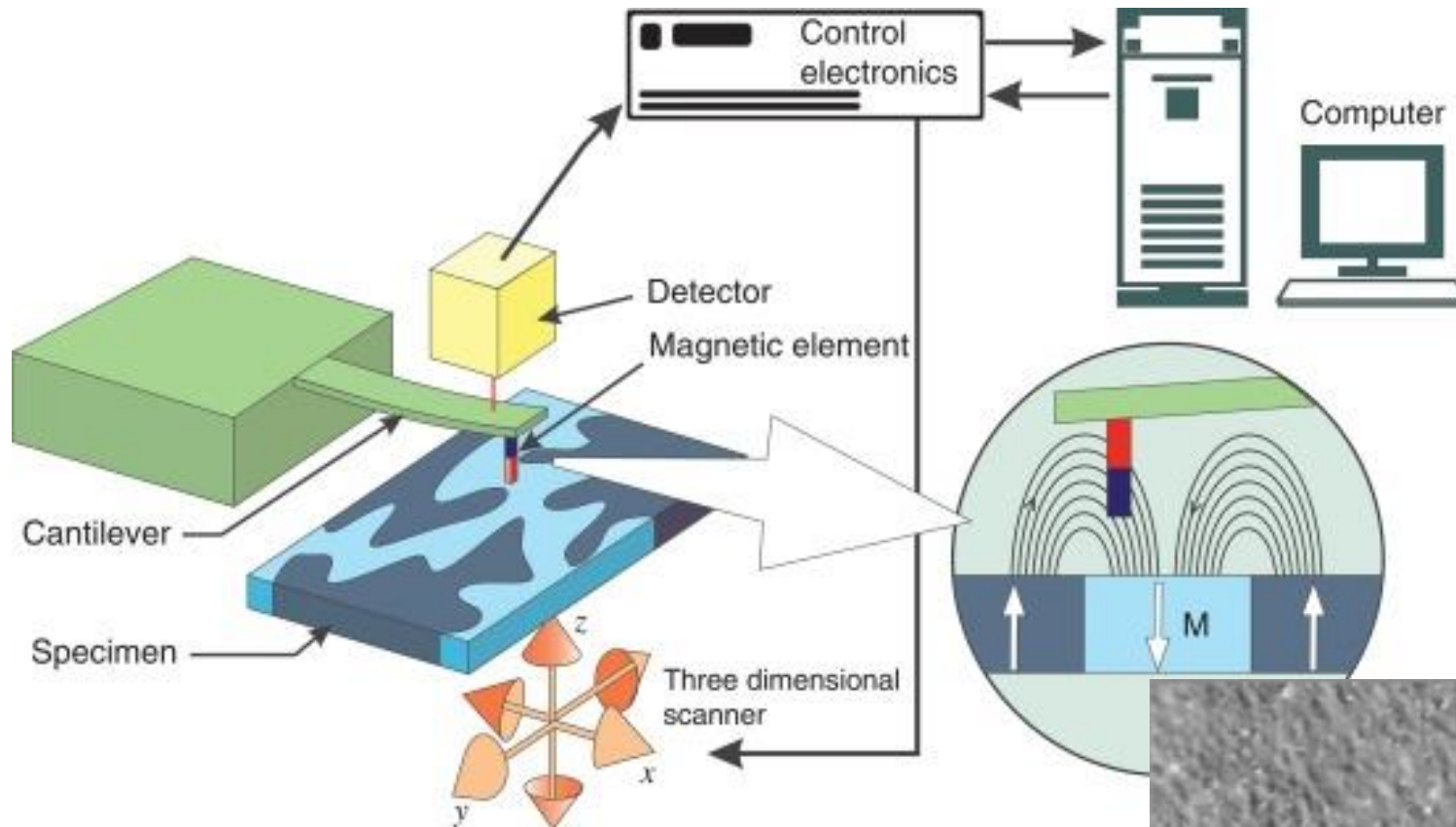


A series of MFM images of the garnet film in the in-plane magnetic field. The field is directed along the X axis; the film was initially in the demagnetized state. (a) $H = 0$ Oe, (b) $H = 35$ Oe, (c) $H = 40$ Oe, (d) $H = 65$ Oe, (e) $H = 205$ Oe, (f) $H = 260$ Oe, (g) $H = 370$ Oe, (h) $H = 470$ Oe, (i) $H = 370$ Oe. The scale bar corresponds to $5 \mu\text{m}$.

**Magnetic Domain Structure
of $\text{Lu}_{2.1}\text{Bi}_{0.9}\text{Fe}_5\text{O}_{12}$**

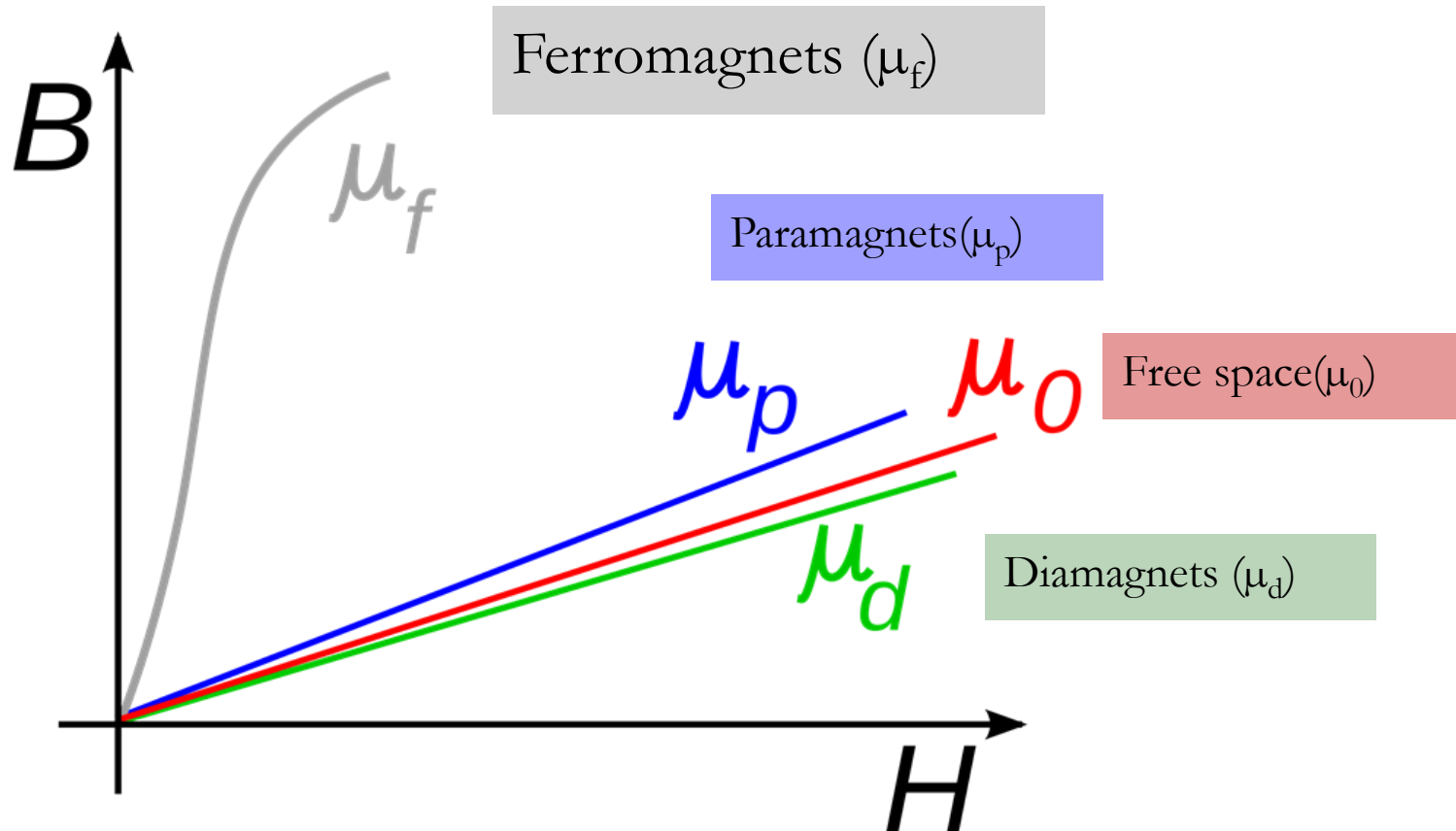
MFM detects magnetic field *gradients*

12 μm $\times$ 12 μm topographic (left) and magnetic force (right) images of a track edge on magnetic recording tape. The magnetic force image was taken at a lift height of 30 nm. The bright and dark regions in the magnetic force image correspond to boundaries between the magnetic bits



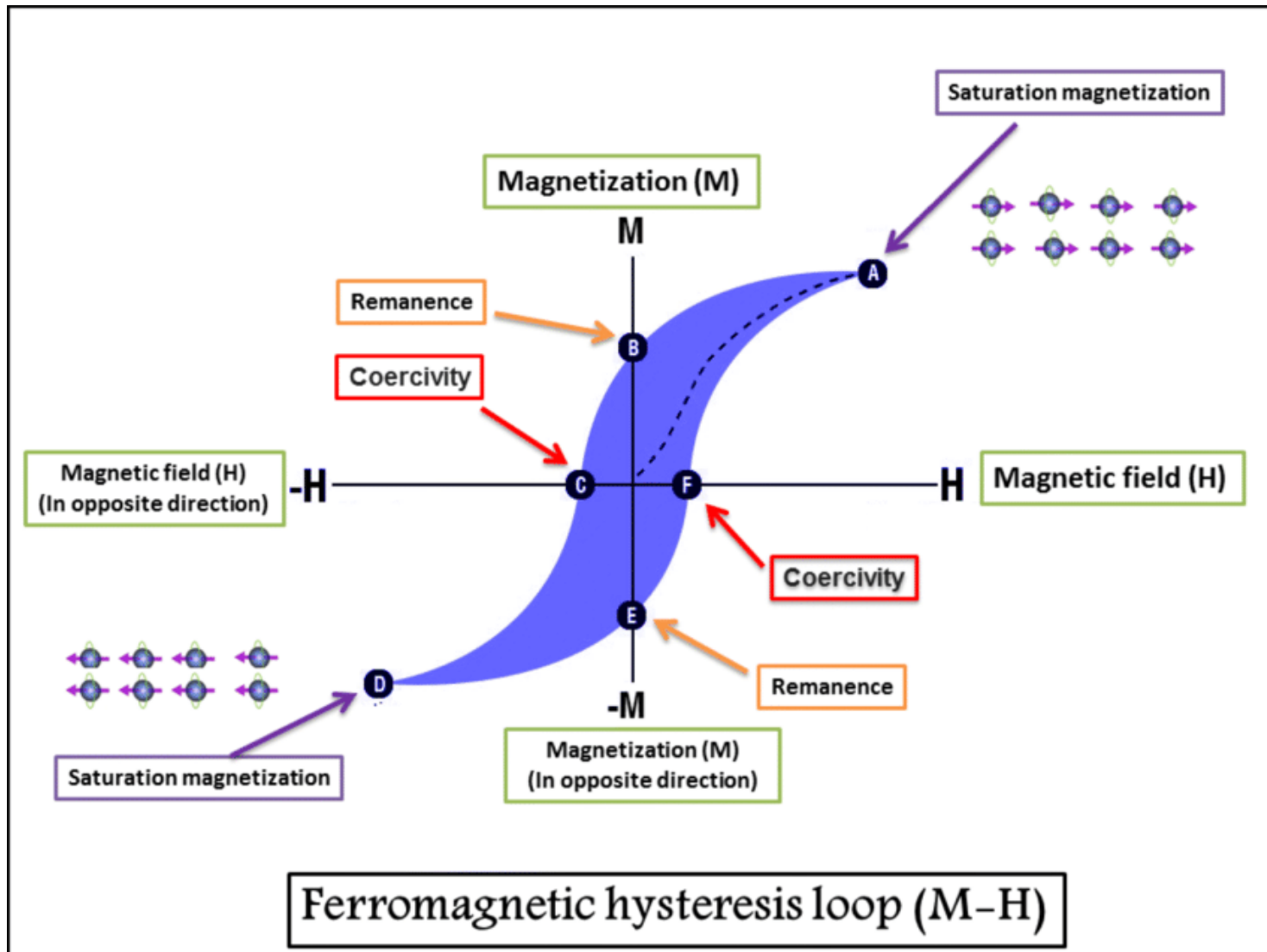
Linear and Nonlinear Media

Comparison of Permeabilities (μ)

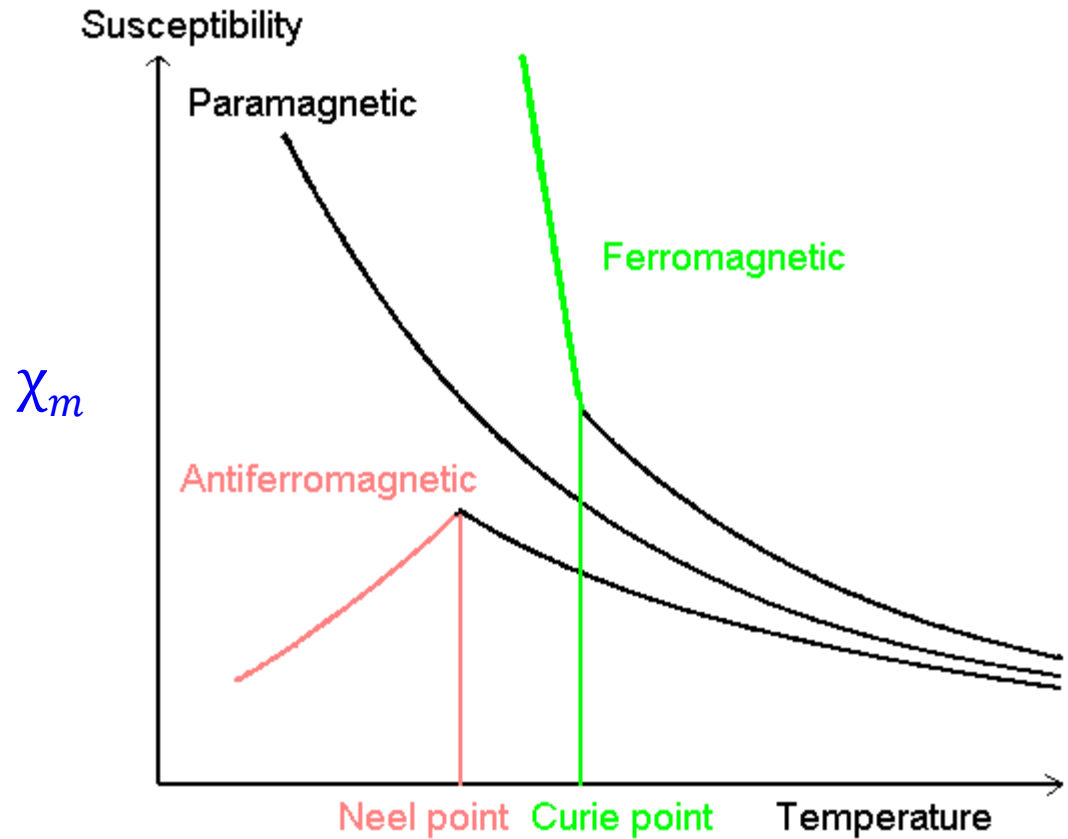


Linear and Nonlinear Media

Comparison of Permeabilities (μ)



Temperature dependence of (χ_m)



Louis Néel

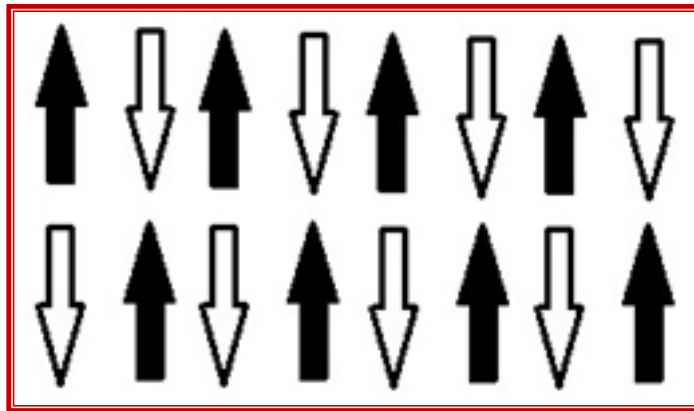


T (Kelvin or °C)

1970 Nobel prize in physics

Antiferromagnetism

In materials that exhibit **antiferromagnetism**, the magnetic moments of atoms or molecules, usually related to the spins of electrons, align in a regular pattern with neighboring spins (on different sublattices) pointing in opposite directions.



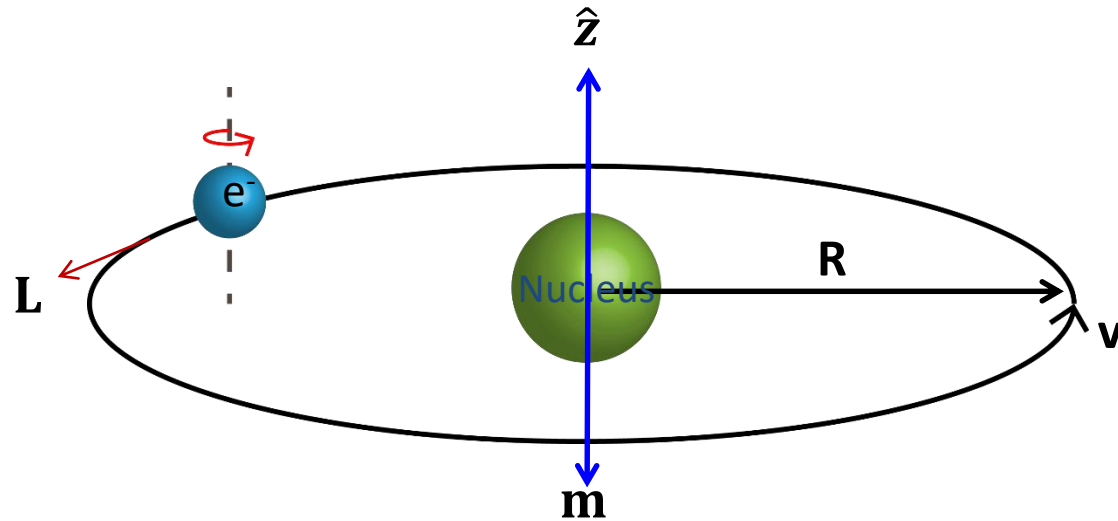
Louis Néel



1970 Nobel prize in physics

Effect of a Magnetic Field on Atomic Orbits

Electrons not only spin they also revolve around the nucleus--for simplicity, let's assume the orbit is a circle of radius R



$$T = \frac{2\pi R}{v}$$

Although technically this orbital motion does not constitute a steady current in practice the period 'T' is so short that unless you is so short that unless you blink awfully fast, it's going to look like a steady

$$I = \frac{e}{T}$$

$$I = \frac{ev}{2\pi R}$$

Accordingly, the orbital dipole moment is

$$m = I\pi R^2$$

$$m = \frac{ev}{2\pi R} \pi R^2$$

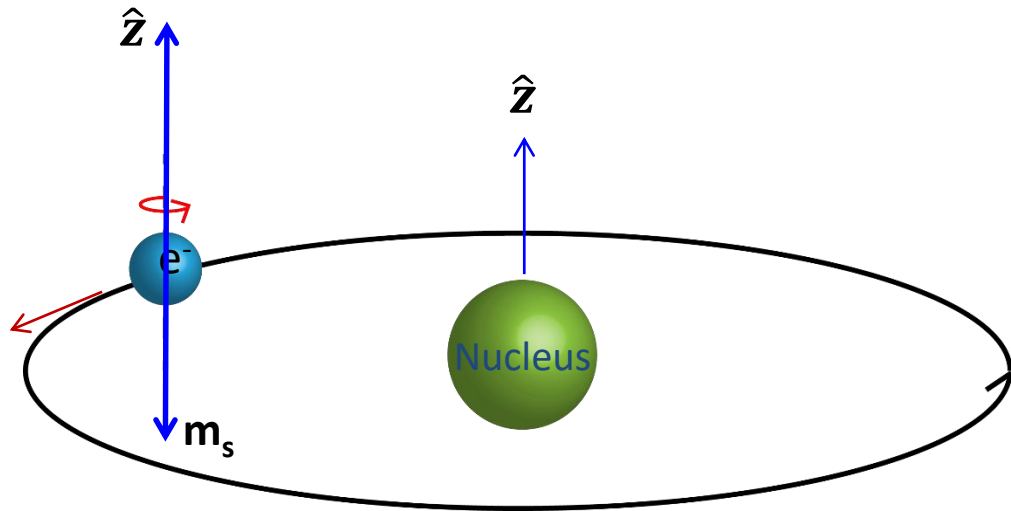
$$m = \frac{e v R}{2}$$

$$\mathbf{m} = -\frac{e v \pi R}{2} \hat{\mathbf{z}}$$

Here, minus sign accounts for the negative charge of the electron

In the presence of Magnetic field it will experience a torque

But it's a lot harder to tilt the entire orbit than it is the spin, so the orbital contribution to Paramagnetism is small.



Instead of tilting the entire orbit, the electron speeds up or slows down depending on the orientation of magnetic field $\mathbf{B}$.

Here, the centripetal acceleration is ordinarily sustained by electrical forces alone.

$$\text{Electrical Coulomb force} \longrightarrow \frac{1}{4\pi \epsilon_0} \frac{e^2}{r^2} = \frac{m_e v^2}{r} \text{ Eq.1} \longleftarrow \text{Centripetal force}$$

In the presence of a magnetic field there is an additional magnetic force

$$\vec{F}_m = -e (\vec{v}' \times \vec{B}) \longleftarrow \text{Magnetic force}$$

For the sake of argument, let's say that B is perpendicular to the plane of the orbit.

$$F_m = ev'B$$

The total force

$$\frac{1}{4\pi \epsilon_0} \frac{e^2}{r^2} + ev'B = \frac{m_e v'^2}{r}$$

Eq.2

Under these conditions, the new speed v' is greater than v by

After subtracting Eq.2 from Eq.1

$$ev'B = \frac{m_e}{r} (v'^2 - v^2)$$

$$ev'B = \frac{m_e}{r} (v' - v)(v' + v)$$

Here, $\Delta v = v' - v$

$$erB = \frac{m_e}{v'} (\Delta v)(v' + v)$$

$$erB = m_e (\Delta v) \left(1 + \frac{v}{v'}\right)$$

$$\frac{v}{v'} \rightarrow 1$$

If Δv is very small

$$e r B = 2 m_e (\Delta v)$$

$$\Delta v = \frac{e r B}{2 m_e}$$

When B is turned on, then, the electron speeds up.

A change in orbital speed means a change in the dipole moment

$$\Delta \mathbf{m} = -\frac{1}{2} e \Delta v R \hat{z}$$

$$\text{since} \\ \mathbf{m} = -\frac{e v \pi R}{2} \hat{z}$$

$$\Delta \mathbf{m} = -\frac{e^2 R^2}{4 m_e} B$$

Note that the change in $\mathbf{m}$ is opposite to the direction of B

- ❑ Ordinarily, the electron orbits are randomly oriented, and the orbital dipole moments cancel out. But in the presence of a magnetic field, each atom picks up a little "extra" dipole moment, and these increments are all antiparallel to the field.
- ❑ This is the mechanism responsible for diamagnetism. It is a universal phenomenon, affecting all atoms. However, it is typically much weaker than paramagnetism, and is therefore observed mainly in **atoms with even numbers of electrons**, where paramagnetism is usually absent.
- ❑ The diamagnetism is a quantum phenomenon not the classical.
- ❑ A complete understanding of these atomic mechanisms requires application of quantum mechanics beyond the scope of this PH-102.

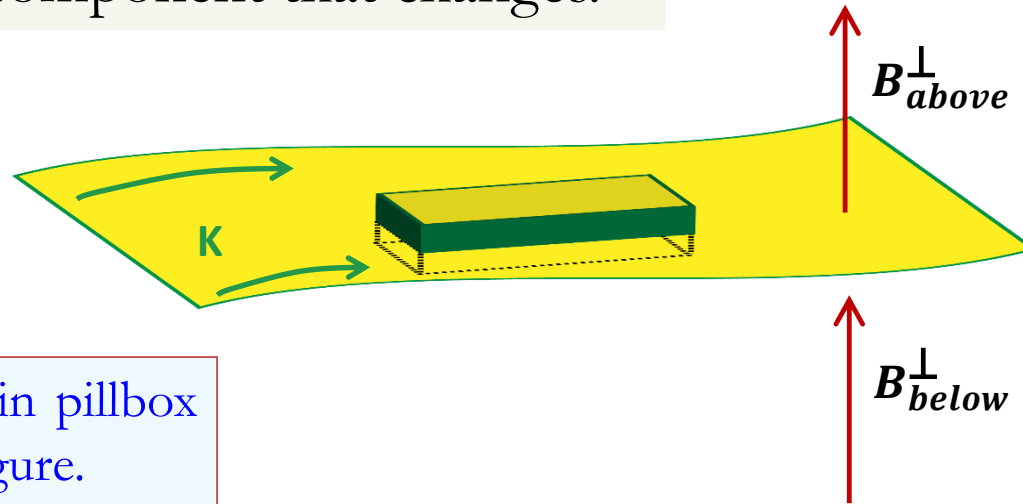
Boundary Conditions

Just as the electric field suffers a discontinuity at a surface charge, so the magnetic field is discontinuous at a surface current.

Only this time it is the tangential component that changes.

$$\oint \vec{B} \cdot d\vec{a} = 0$$

If we apply this equation to a wafer thin pillbox straddling the surface as shown in this figure.



This Gaussian pillbox is vertically centered on the interface between the two magnetic media. We then shrink the height of pillbox to infinitesimally above/below the interface, then only the top/bottom portions of the surface integral will contribute anything.

We thus obtain the condition:

$$B_{above}^\perp = B_{below}^\perp$$

We also have the constitutive relation

$$\frac{1}{\mu_0} \vec{B} = \vec{H} + \vec{M}$$

$$\frac{1}{\mu_0} \nabla \cdot \vec{B} = \nabla \cdot \vec{H} + \nabla \cdot \vec{M}$$

$$\nabla \cdot \vec{H} = -\nabla \cdot \vec{M}$$

Since

$$\nabla \cdot \vec{B} = 0$$

In integral form this relation becomes

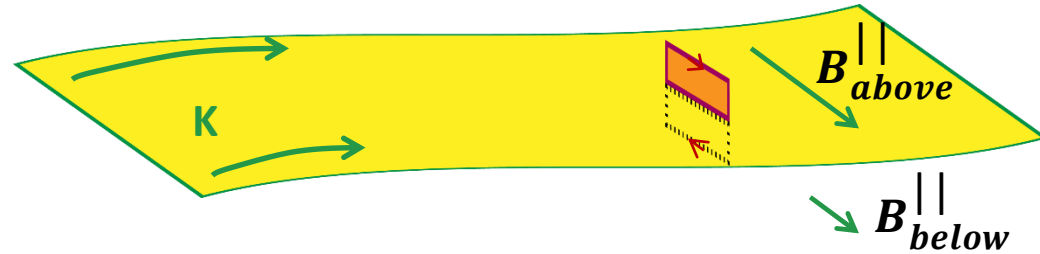
$$\oint_S \vec{H} \cdot d\vec{a} = - \oint_S \vec{M} \cdot d\vec{a}$$

Using the same Gaussian pillbox, we obtain the following condition on the perpendicular components of $\vec{H}$ and $\vec{M}$ above/below interface:

$$\left[H_{\text{above}}^{\perp} - H_{\text{below}}^{\perp} \right]_{\text{surface}} = - \left[M_{\text{above}}^{\perp} - M_{\text{below}}^{\perp} \right]_{\text{surface}}$$

As for the tangential components, an Amperian loop running perpendicular to the current.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$



We can take a (rectangular) contour vertically centered above/below the interface, we then shrink the height of this contour to be infinitesimally above/below the interface, thus only the tangential portions of the line integral above/below the interface will contribute.

$$\left(B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} \right) l = \mu_0 I_{\text{enclosed}}$$

$$\left(B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} \right) l = \mu_0 K l$$

$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$$

Thus the component of $\mathbf{B}$ that is parallel to the surface but perpendicular to the current is discontinuous in the amount $\mu_0 K$.

A similar Amperian loop running parallel to the current reveals that the parallel component is continuous. These results can be summarized in a single formula:

$$\vec{B}_{\text{above}} - \vec{B}_{\text{below}} = \mu_0 (\vec{K} \times \hat{n})$$

Here $\hat{n}$ is a unit vector perpendicular to the surface, pointing "upward."

Like the scalar potential in electrostatics, the vector potential is continuous across the boundary:

$$\vec{A}_{\text{above}} = \vec{A}_{\text{below}}$$

The $\nabla \cdot \vec{A} = 0$ guaranties that the normal component is continuous.

$$\nabla \times \vec{A} = \vec{B}$$

In the integral form

$$\oint \vec{A} \cdot d\vec{l} = \oint \vec{B} \cdot d\vec{a} = \Phi$$

The tangential components are continuous (the flux through an Amperian loop of vanishing thickness is zero).

Since

$$\nabla \times \vec{A} = \vec{B}$$

This relation can also be equivalently written as

$$\frac{\partial A_{above}}{\partial n} - \frac{\partial A_{below}}{\partial n} = -\mu_0 \vec{K}$$

Similarly, Ampere's Law for $\vec{H}$

Differential Form

$$\nabla \times \vec{H} = \vec{J}_f$$

$$H_{above}^{\parallel} - H_{below}^{\parallel} = K_{free}$$

Integral Form

$$\oint \vec{H} \cdot d\vec{l} = I_{f,enclosed}$$

$$\vec{H}_{above} - \vec{H}_{below} = \vec{K}_{free} \times \hat{n}$$

Again, we can take a (rectangular) contour vertically centered above/below the interface between the two magnetic media; we then shrink the height of this contour to be infinitesimally above/below the interface, thus only the tangential portions of the line integral above/below the interface will contribute.

Linear and Nonlinear Media

Magnetic Susceptibility and Permeability

In paramagnetic and diamagnetic materials, the magnetization is sustained by the field; when $\mathbf{B}$ is removed, $\mathbf{M}$ disappears.

In fact, for most substances the magnetization is proportional to the field, provided the field is not too strong.

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

For Electrical case

If we follow the electrical analogy we can write the proportionality as:

$$\vec{M} = \frac{1}{\mu_0} \chi_m \vec{B}$$

Incorrect!!!

Custom dictates that it be written in terms of H , instead of B :

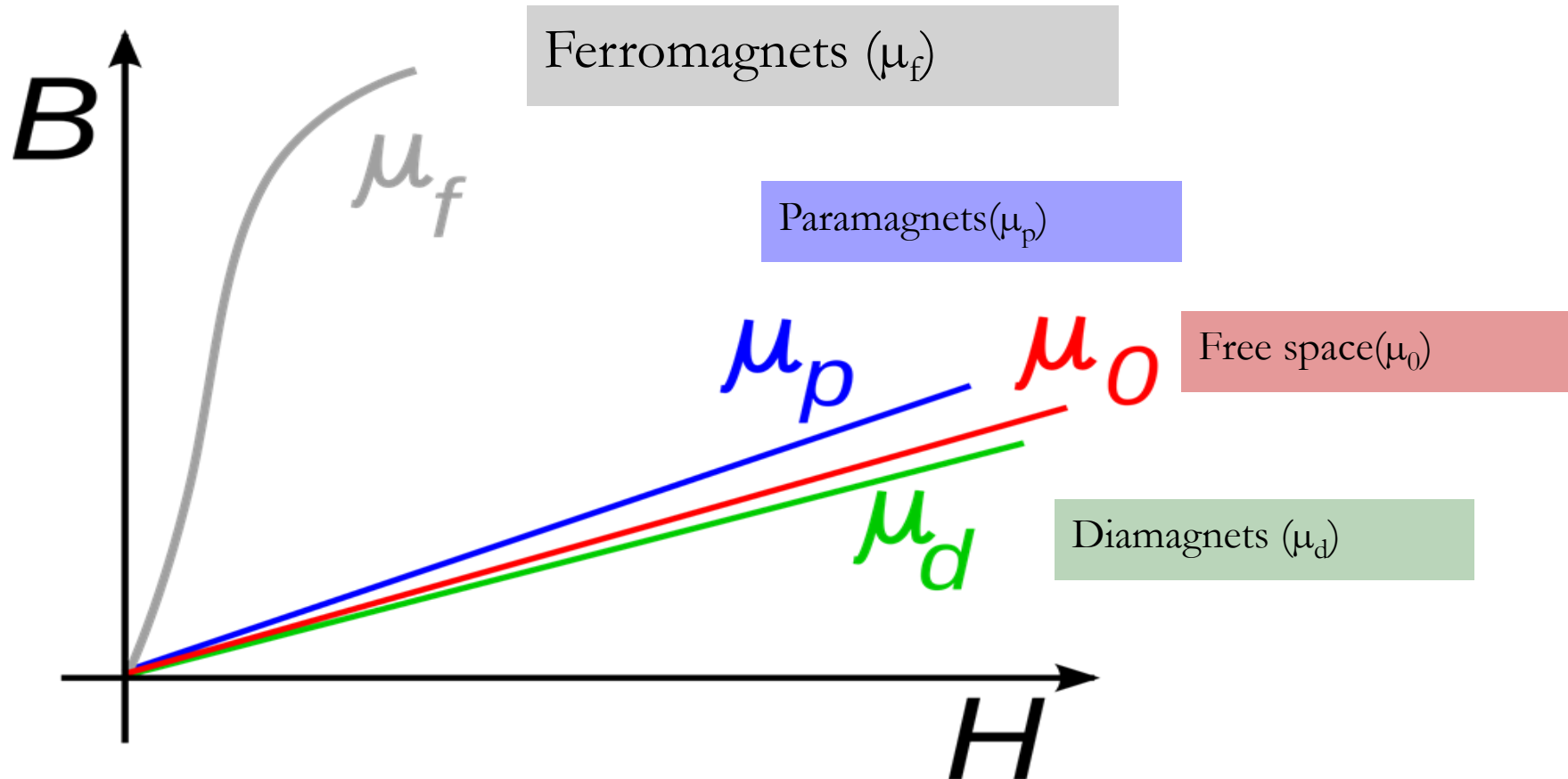
$$\vec{M} = \chi_m \vec{H}$$

Equation 1

The constant of proportionality χ_m is called the magnetic susceptibility

Nonlinearity and Law of approach to saturation magnetization

It has been experimentally confirmed that ferromagnetic materials follow the law of approach to saturation: $\mathbf{M} = M_s (1 - a/H - b/H^2) + \chi H$ where M_s is the saturation magnetization and a , b and χ are constants.



Few important points:

- ❑ χ_m is a dimensionless quantity that varies from one substance to another.
- ❑ It is positive for paramagnets and negative for diamagnets.

Materials that obey Equation 1 are called linear magnetic materials.

$$\vec{B} = \mu_0(\vec{H} + \vec{M})$$

$$\vec{B} = \mu_0(\vec{H} + \chi_m \vec{H})$$

$$\vec{B} = \mu_0(1 + \chi_m)\vec{H}$$

Thus B is also proportional to H (for linear magnetic materials):

$$\vec{B} = \mu \vec{H}$$

$$\mu = \mu_0(1 + \chi_m)$$

Where

μ is called the permeability of the material.

In a vacuum, where there is no matter to magnetize, the susceptibility χ_m vanishes, and the permeability is μ_0 . That's why μ_0 is called the permeability of free space.

Dimensions of μ_0 (permeability of free space).

μ_0 is actually defined as

$$\mu_0 = \frac{1}{\epsilon_0 c^2}$$

In electrostatics, the electric permittivity of free space ϵ_0 is 8.85×10^{-12} Farads/meter

Thus $\mu_0 = 4\pi \times 10^{-7}$ Newton/Ampere²

kg – meter/(sec Ampere)²

Henry/meter

Summary

Four Different Kinds of Magnetism

Types of
Magnetism

```
graph TD; A[Types of Magnetism] --> B[Diamagnetism]; A --> C[Paramagnetism]; A --> D[Ferromagnetism]; A --> E[Anti-ferromagnetism]; B --> B1[The magnetization is in the opposite direction to that of the applied field]; C --> C1[Positive (but small) susceptibility]; D --> D1[Bulk magnetic moment and the magnetization is large]; E --> E1[Anti-parallel alignment of spins (Total spontaneous magnetization is zero)];
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Diamagnetism

The magnetization is in the opposite direction to that of the applied field

Paramagnetism

Positive (but small) susceptibility

Ferromagnetism

Bulk magnetic moment and the magnetization is large

Anti-ferromagnetism

Anti-parallel alignment of spins (Total spontaneous magnetization is zero)

Electromagnetic Induction (General Introduction)

Michael Faraday was one of the fathers of modern electricity and went on to influence and inspire other brilliant minds, including Albert Einstein.

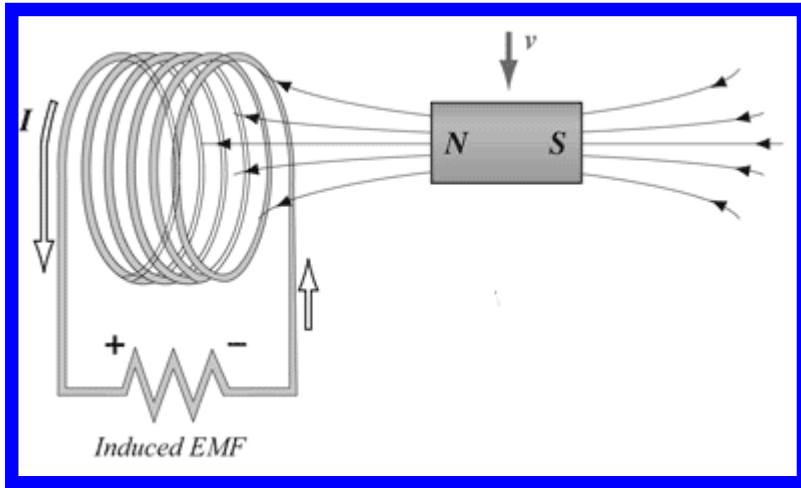


“Nothing is too wonderful to be true, if it be consistent with the laws of nature.” – **Michael Faraday**

In 1831 Michael Faraday reported on a series of experiments

Faraday's Law

General Introduction

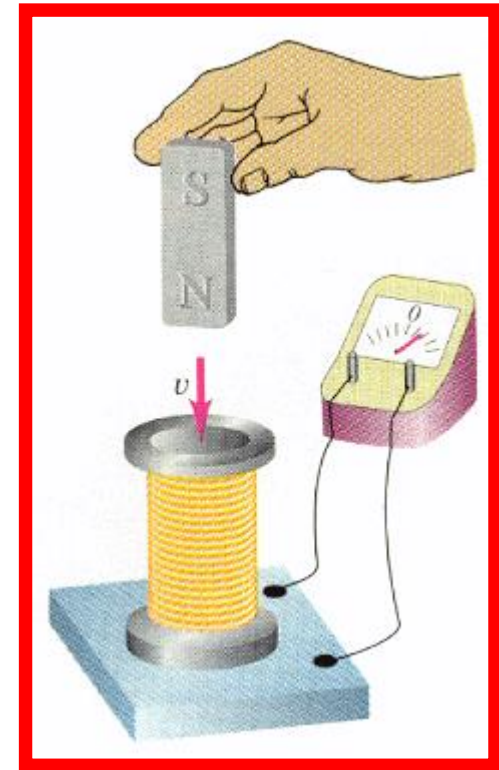


$$\varepsilon = - \frac{\Delta \Phi_m}{\Delta t}$$

$\frac{\Delta \Phi_m}{\Delta t}$ = Rate of change of magnetic flux through the circuit

ε = Induced Electromotive Force (emf)

Lenz's law: According to this law, the direction of induced emf or current in a circuit is such as to oppose the cause that produces it. This law gives the direction of induced emf/induced current. This law is based upon law of conservation of energy.



Heinrich Lenz

Note: This slide gives just a brief introduction we will study this effect in detail later

Electromotive Force (emf)

General Introduction

To make a current flow, you have to push on the charges. How fast they move, in response to a given push, depends on the nature of the material. For most substances, the current density $\mathbf{J}$ is proportional to the force per unit charge, $\mathbf{f}$:

$$\mathbf{J} = \sigma \mathbf{f}$$

Equation -1

$$\mathbf{J} \propto \mathbf{E}$$

$$\mathbf{J} = \sigma \mathbf{E} = \mathbf{E} / \rho$$

The proportionality factor (not to be confused with surface charge) is an empirical constant that varies from one material to another; it's called the conductivity of the medium.

Resistivity is defined as

$$\rho = \frac{1}{\sigma}$$

In fact, for most purposes metals can be regarded as perfect conductors, with $\sigma = \infty$.

In principle, the force that drives the charges to produce the current could be anything chemical, gravitational, or trained ants with tiny harnesses.

Electric Current: Analogy, water flowing in a pipe

H_2O
gallons/minute



“Flow Rate” is the NET amount of water passing through a surface per unit time

Individual molecules are bouncing around with speeds of km/s!

Net water velocity is m/s

I
Coulombs/s



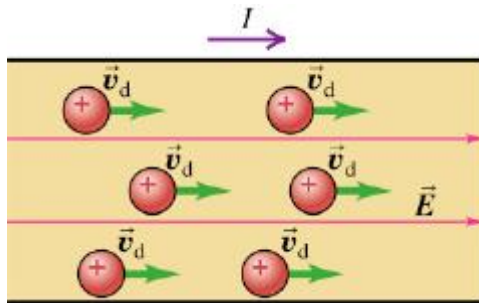
“Electric Current” is the NET amount of charge passing through a surface per unit time

Individual electrons are bouncing around with very high speed

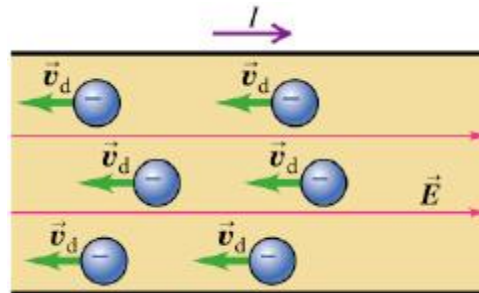
Electron “drift velocity” may be mm/s

Note: Propagation speed (V_p) is different than V_d
($V_p \gg V_d$)

In a Conductor, Charges are free to move.



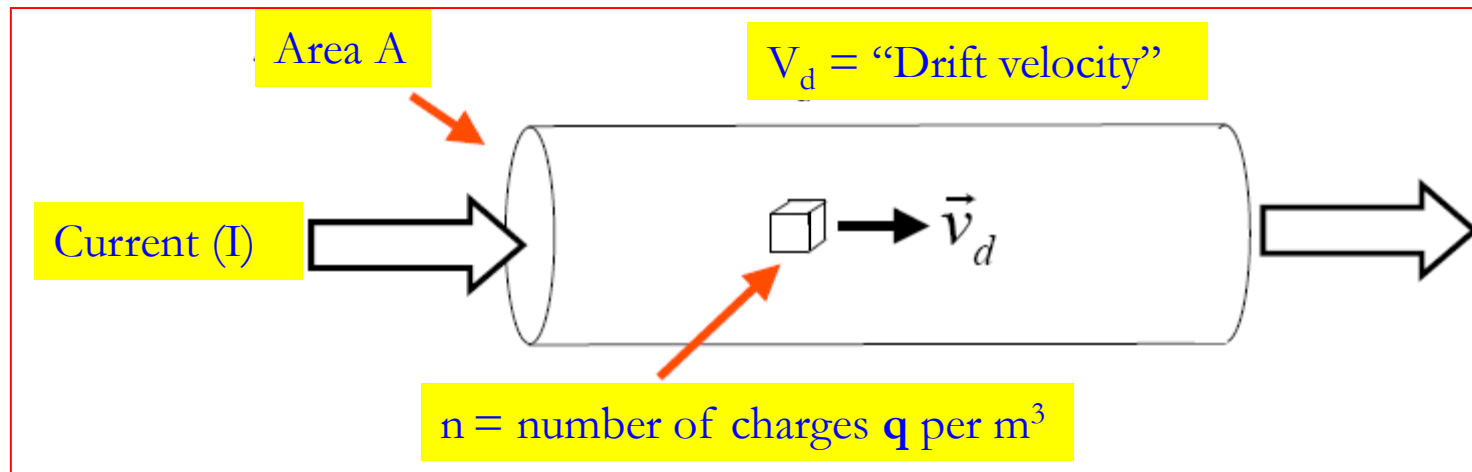
The charges may be positive; This is usually relevant only for “special cases” like ions in a solution. (Holes in semiconductors act like positive charges)



The charges may be negative; This is the normal case for metallic conductors

$$I = \frac{dQ}{dt}$$

Inside a conductor there are lots of charges
There could be 10^{24} electrons /cm²



Total Current through area **A** is given by

$$I = nqv_d A$$

Current per unit area is given by

$$J = \frac{I}{A} = nqv_d$$

Note: Propagation speed (V_p)
is different than V_d
($V_p \gg V_d$)

$$\vec{J} = nq\vec{v}_d$$

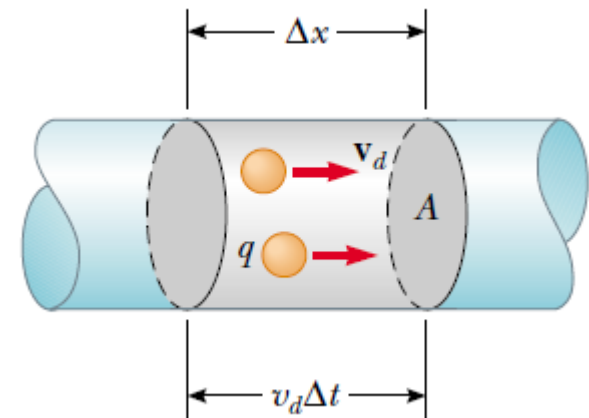
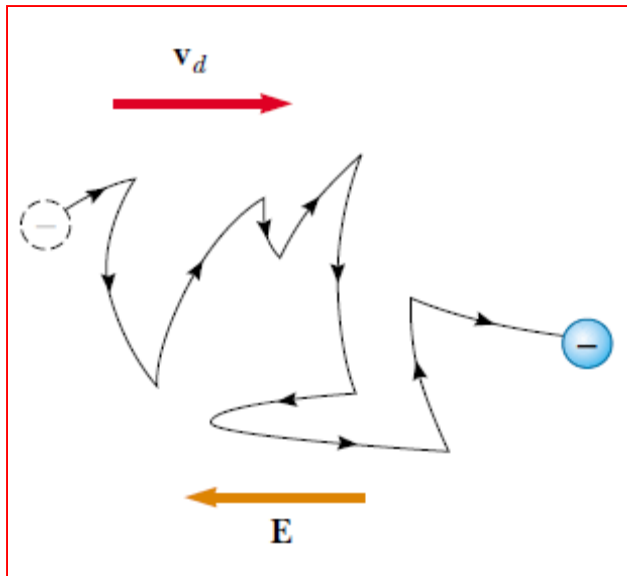


Vector Current Density

J can vary in magnitude and direction in Space

Drift velocity (V_d)

A schematic representation of the zigzag motion of an electron in a conductor. The changes in direction are the result of collisions between the electron and atoms in the conductor. Note that the net motion of the electron is opposite the direction of the electric field. Each section of the zigzag path is a parabolic segment.



$$\Delta Q = (nAv_d \Delta t) q$$

$$I_{av} = \frac{\Delta Q}{\Delta t} = nqv_d A$$

Material	Resistivity (Ω)	Material	Resistivity (Ωm)
Conductors		Semiconductors	
Silver	1.59×10^{-8}	Salt Water(saturated)	4.4×10^{-2}
Copper	1.68×10^{-8}	Germanium	4.6×10^{-1}
Gold	2.21×10^{-8}		
Aluminum	2.65×10^{-8}	Silicon	2.5×10^3
Iron	9.61×10^{-8}	Insulators	
Mercury	9.58×10^{-7}	Water (pure)	2.5×10^5
Nichrome	1.00×10^{-6}	Wood	$10^8 - 10^{11}$
Manganese	1.44×10^{-6}	Glass	$10^{10} - 10^{14}$
Graphite	1.4×10^{-5}	Quartz (fused)	10^{16}

Table 1: Resistivity, in Ohm (Ω)-meters (all values are for 1 atm, 20° C).

Source: Handbook of Chemistry and Physics, 78th ed.(Boca Raton: CRC Press, Inc., 1997).

Understanding the Units

$$I = \frac{dQ}{d\tau} \text{ (Amperes)}$$

$$1 \text{ Ampere} = 1 \text{ Coulomb/sec}$$

SI units of electrical conductivity σ_c :

Volume free current density: $\vec{J}_{free}(\vec{r})$ Amperes/meter² (A/m^2)

Force per unit charge: $\vec{f}(\vec{r}) \equiv \frac{\vec{F}(\vec{r})}{Q}$ Newtons/Coulomb (N/C)

$$\sigma_c = \left| \frac{\vec{J}(\vec{r})}{\vec{f}(\vec{r})} \right| = \frac{A/m^2}{N/C} = \underline{\underline{C^2/N \cdot m^2 \cdot s}}$$

But: $\vec{f}(\vec{r}) = \frac{\vec{F}(\vec{r})}{Q} \left(\frac{N}{C} \right)$ also has SI units of: $\vec{E}(\vec{r}) = \frac{\vec{F}(\vec{r})}{Q} \left(\frac{\text{Newtons}}{\text{Coulomb}} = \frac{\text{Volts}}{\text{meter}} \right)$

$$\sigma_c = \frac{\text{Amps}/m^2}{\text{Volts}/m} = \left(\frac{\text{Amps}}{\text{Volt}} \right) / m$$

Define resistance: $R \equiv \frac{|\Delta V|}{I} = \frac{|V_2 - V_1|}{I} \left(\frac{\text{Volts}}{\text{Amps}} \equiv \text{Ohms} \right)$

SI units symbol = Ω

⇒ SI units of electrical conductivity:

$\sigma_c = \text{Ohms}^{-1}/\text{m}$	$1 \Omega \equiv 1 \text{ V} / \text{A}$
	$1 \text{ Ohm} \equiv 1 \text{ Volt} / \text{Amp}$

Define electrical resistivity of material:

$\rho_c \equiv \frac{1}{\sigma_c}$	SI units of $\rho_c = \text{Ohm-m}$
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$1 \text{ Ohm}^{-1} \equiv \text{“mho”}$ (Ohm spelled backwards)

$\sigma_c = \text{ohm}^{-1} / \text{m} = \text{mho} / \text{m}$

Note also that (esp. in Europe):

$1 \text{ Siemens} \equiv 1 \text{ Ampere} / \text{Volt}$

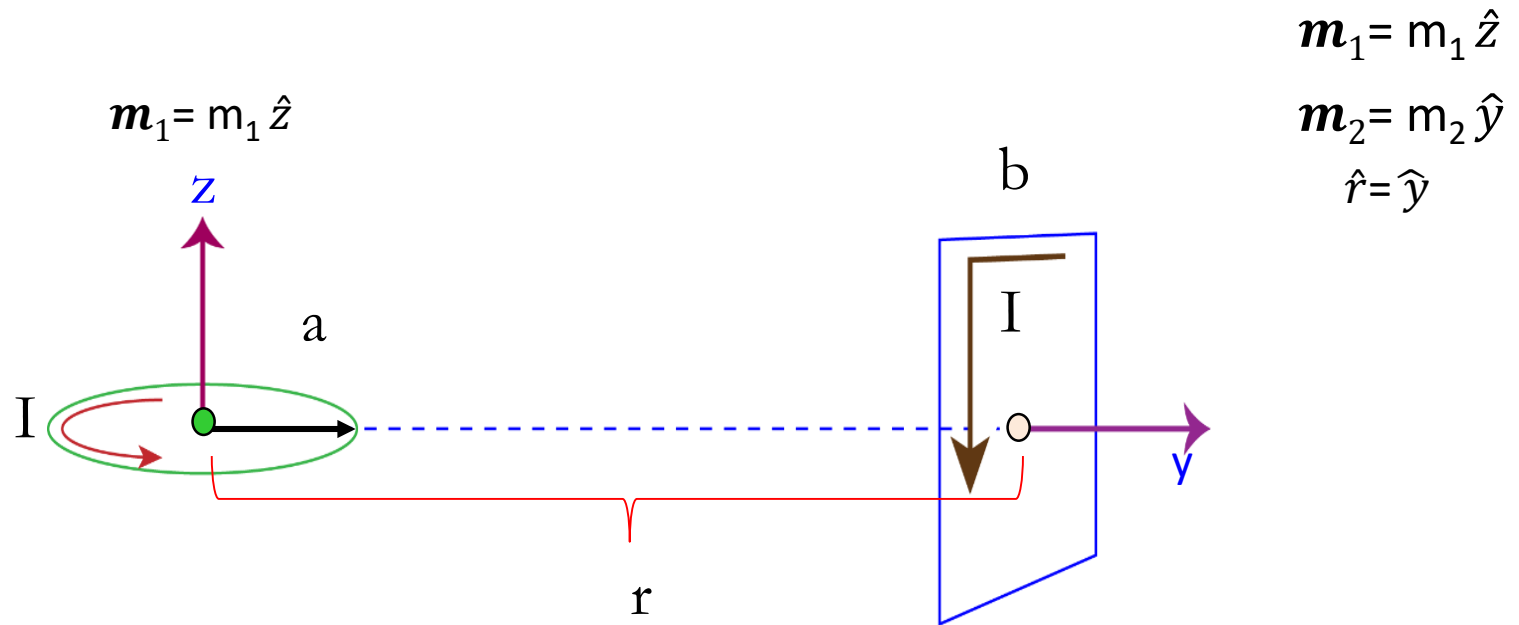
∴ SI units of electrical conductivity:

$\sigma_c = \left(\frac{\text{Amps}}{\text{Volt}} \right) / \text{m} = \text{Siemens/m}$

$\therefore 1 \text{ Siemens} \equiv 1 / \text{Ohm} = 1 \text{ mho}$
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Example:

Calculate the Torque exerted on the square loop shown in the below figure, due to the circular loop (assume r is much larger than a or b). If the square loop is free to rotate, what will its equilibrium orientation be?

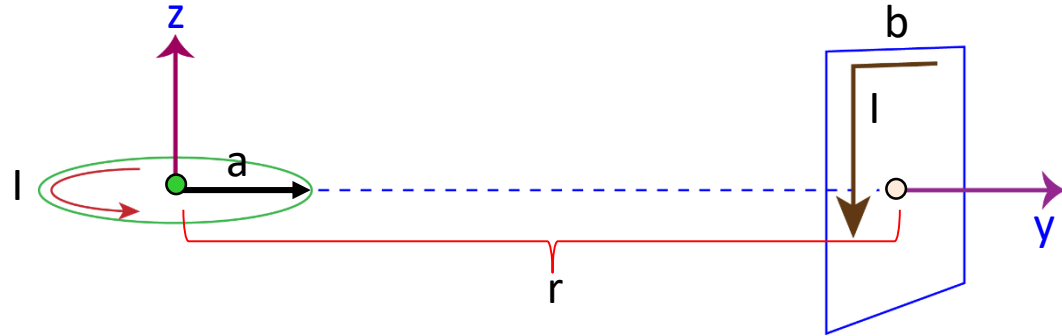


Solution 1:

Torque

$$\vec{N} = \vec{m} \times \vec{B}$$

The magnetic field 'B' at a point 'r' due to a dipole having magnetic moment 'm'



$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{r})\hat{r} - \mathbf{m}]$$

Using this formula, the magnetic field 'B' at the center of square loop, due to the circular coil is

$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m}_1 \cdot \hat{r})\hat{r} - \mathbf{m}_1]$$

$$\begin{aligned} \mathbf{m}_1 &= m_1 \hat{z} \\ \mathbf{m}_2 &= m_2 \hat{y} \\ \hat{r} &= \hat{y} \end{aligned}$$

$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(m_1 \hat{z} \cdot \hat{y})\hat{y} - m_1 \hat{z}]$$

$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{1}{r^3} [-m_1 \hat{z}]$$

$$\hat{z} \cdot \hat{y} = 0$$

$$\mathbf{B}_1 = -\frac{\mu_0}{4\pi} \frac{1}{r^3} m_1 \hat{z}$$

Torque on the square loop

$$\vec{N} = \vec{m}_2 \times \vec{B}_1$$

$$\vec{N} = m_2 \hat{y} \times -\frac{\mu_0}{4\pi} \frac{1}{r^3} m_1 \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

$$\vec{N} = -\frac{\mu_0}{4\pi} \frac{1}{r^3} m_1 m_2 \hat{x}$$

Here

$$m_1 = \pi a^2 I$$

$$m_2 = b^2 I$$

$$\vec{N} = -\frac{\mu_0}{4\pi} \frac{1}{r^3} \pi a^2 b^2 I^2 \hat{x}$$

$$\vec{N} = -\frac{\mu_0}{4\pi} \frac{(abI)^2}{r^3} \hat{x}$$

If the square loop is free to rotate, its equilibrium orientation will be in the direction of applied external magnetic field i.e. Magnetic field of the circular coil at the square loop.

$$\mathbf{B}_1 = -\frac{\mu_0}{4\pi} \frac{1}{r^3} m_1 \hat{z}$$



Downward (negative z-direction)